HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged **low-level device drivers** for hardware components including **GPIO**, **interrupt controllers**, and **voltage regulators**, working directly with hardware schematics.
- Collaborated with hardware design engineers to define and refine **hardware/software interfaces** for new SoC architectures, ensuring seamless integration of drivers.
- Authored **Python scripts** to automate repetitive tasks in the driver development lifecycle, including test case generation and log analysis, reducing manual effort by approximately **30%**.
- Contributed to the development of **C++** code for embedded systems, focusing on performance optimization and memory management within resource-constrained environments.
- Implemented and tested **ARM architecture** specific code segments, ensuring compatibility and efficiency across different processor cores.
- Managed driver version control and code reviews using **Git**, maintaining code quality and facilitating team collaboration within an Agile framework.
- Containerized driver development and testing environments using **Docker**, enabling consistent and reproducible builds across developer workstations and CI/CD pipelines.

SKILLS

Languages – C, C++, Python, Java, SQL, R

Embedded Systems – Device Drivers, Low-Level Programming, ARM, RISC-V, SoC, GPIO, Interrupts, Power Management, JTAG, Lauterbach

Tools & Technologies – Docker, Git, Jenkins, Jira, Confluence, Splunk, Dynatrace, CloudWatch

Databases – PostgreSQL, MySQL, Oracle

Cloud Platforms – AWS (EC2, S3, Lambda)

Scripting & Automation – Python Scripting, Shell Scripting

PROJECTS

Embedded Linux Driver Development for IoT Devices

- Developed a custom **device driver** in **C** for managing sensor data acquisition on an embedded Linux platform, integrating with the system's **interrupt handling** mechanism.
- Created **Python automation scripts** to streamline the testing process for the driver, including data validation and performance benchmarking against established baselines.
- Containerized the driver build and testing environment using **Docker**, ensuring consistent deployment across different developer machines and CI/CD stages.

SoC Configuration and Control Automation

- Engineered a suite of **Python scripts** to automate the configuration of **System on Chip (SoC)** parameters, including **clock speeds** and **voltage regulators**, via a hardware abstraction layer.
- Integrated **low-level hardware control** functionalities into a central automation framework, reducing manual configuration time for new hardware revisions by an estimated **40%**.
- Wrote detailed documentation and provided knowledge transfer sessions to the operations team on the usage and maintenance of the automation tooling.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)